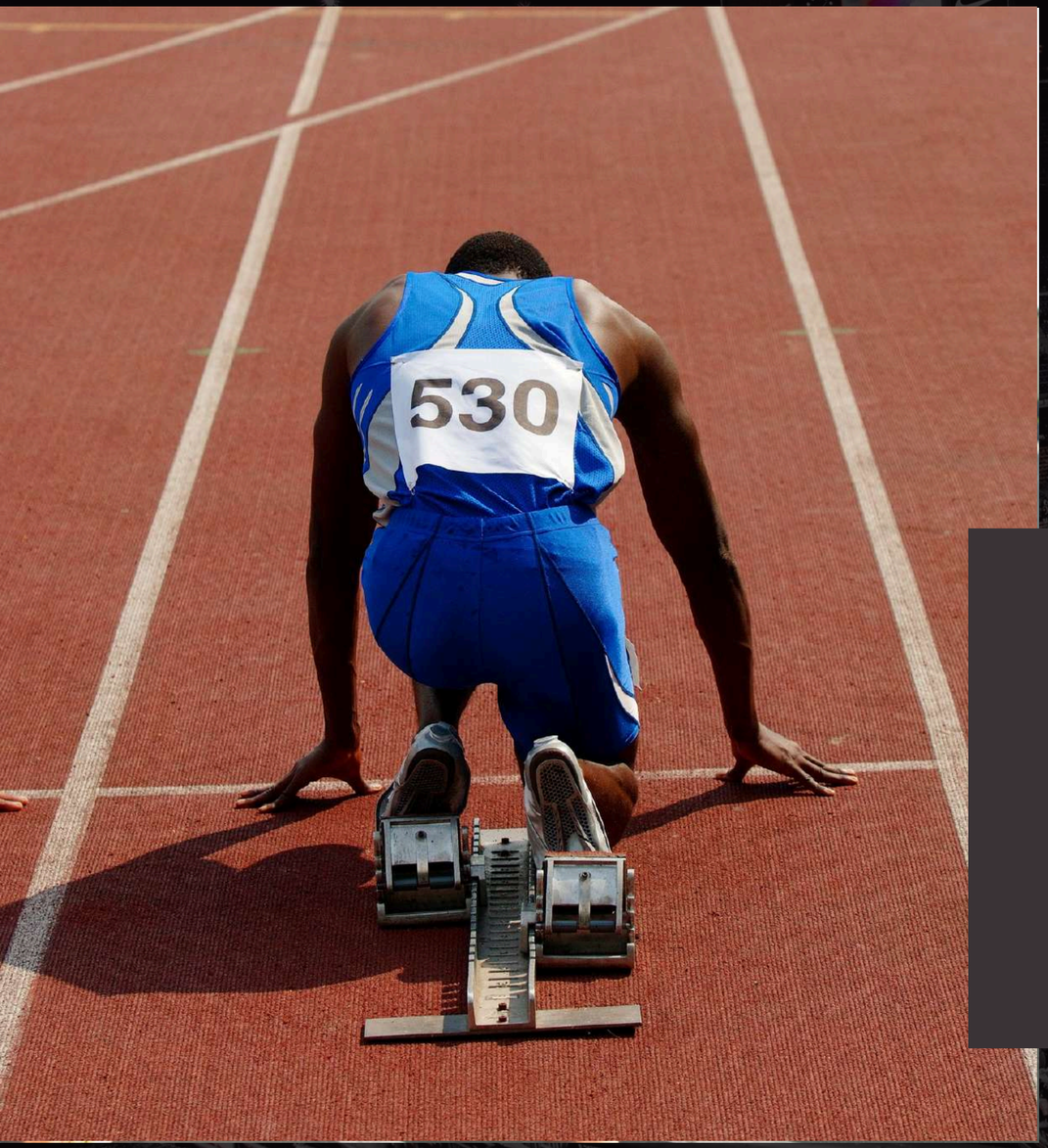


AI ATHLETIC COPILOT PROJECT

Production-Ready LLMOps System



What is project?

Objective: An end-to-end intelligent athletic copilot system that builds structured training, nutrition, and recovery plans powered by grounded knowledge bases.

Problem and solution



The Problem

Unstructured Output: Standard AI assistants return long paragraphs that are difficult for athletes and coaches to parse quickly during training.

Lack of Grounded Data: Generic models often hallucinate nutrition guidelines or workout split recommendations without referencing validated athletic science.

The Solution

AI Athletic Copilot: An end-to-end, LLMOps-driven assistant tailored for sports performance.

Structured UI: Formats workouts, track intervals, and nutrition macros directly into clean Markdown tables and UI metrics

Domain Grounding: Integrates a Retrieval-Augmented Generation (RAG) framework referencing validated ISSN guidelines.

Core LLMOps Architecture

1. **Hybrid RAG System** (Retrieval-Augmented Generation)

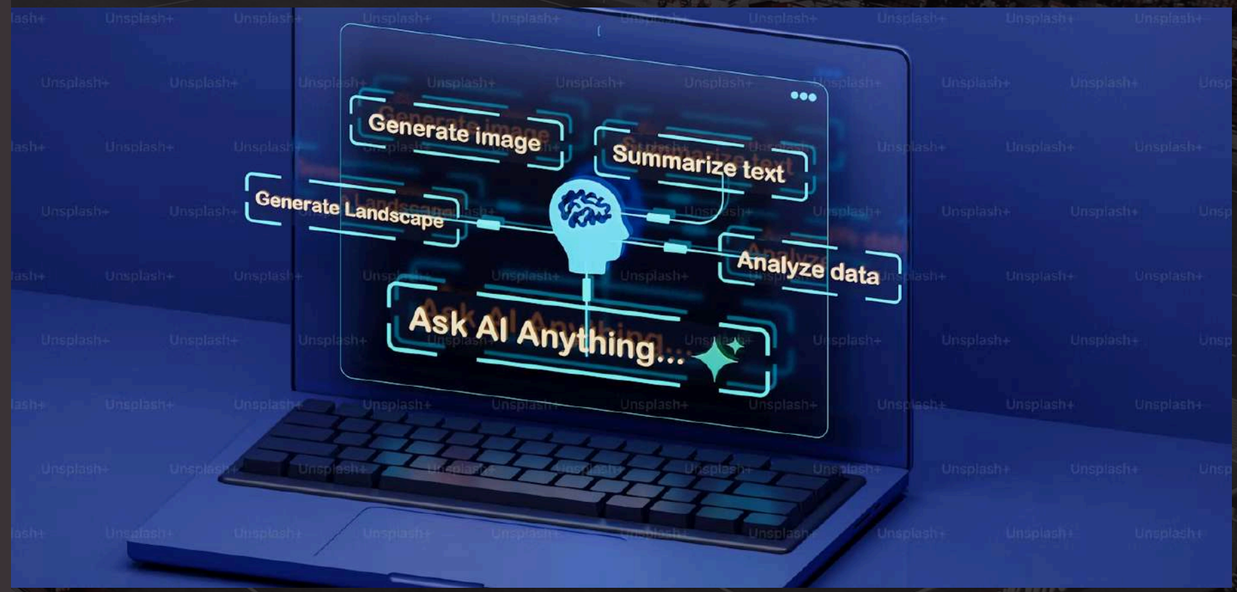
Persistent Knowledge Base: Reads from a local athletic_knowledge_base.txt storing ISSN sports science standards.

Dynamic Document Ingestion: Allows athletes to upload custom PDF files (e.g., blood work, medical tests, specific track plans) for real-time contextual integration using pypdf.

2. **Strict Output Parsing** (Pydantic & JSON Output Parsers)

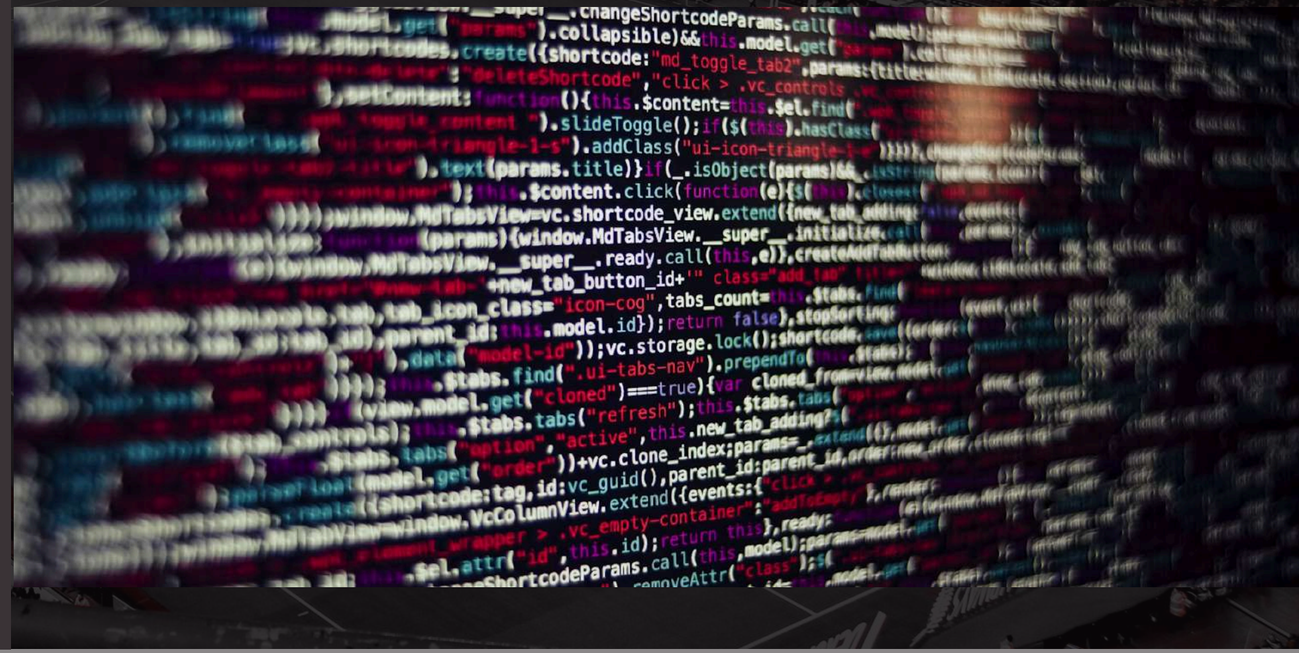
Enforces strict data models via Pydantic Schemas (NutritionScheduleSchema and WorkoutSchema).

Leverages JsonOutputParser to eliminate unstructured text responses and populate native Streamlit components cleanly.



Core LLMOps Architecture

3. **Contextual Memory & System Prompt Engineering**
 Maintains conversation context across user interactions using Streamlit's session state.
 Leverages a tailored System Prompt that automatically structures track running programs into defined warm-up, main session, rest interval, and cool-down columns.
4. **Input Guardrails & Domain Safety**
 Features a pre-execution validation layer to verify that queries fall strictly within the athletic, sports, nutrition, and recovery domains.





Application Features & User Interface

🥗 RAG Nutrition Planner:

Calculates daily target calories and macro splits while displaying retrieved scientific context alongside customized meal plans.

🏋️ Custom Gym Program Generator:

Builds targeted resistance training programs based on athlete experience, frequency, available equipment, and hyper-specific performance goals.

💧 Hydration & Recovery Engine:

Dynamically computes daily fluid intake requirements based on body mass and session duration, alongside recovery protocols.

💬 Structured AI Coach Chat:

Fixed-bottom conversational interface designed for rapid query response, featuring clear history triggers and table output rendering.

Tech Stack & Tools

Frontend & UI Framework: Streamlit

LLM Core Engine: gpt-4o (via g4f.client)

Framework & Parsing: LangChain Core (JsonOutputParser) & Pydantic

Document Processing: PyPDF **(**PdfReader)

Storage & State: Native Python File I/O & Streamlit Session State





Production Readiness & Live Demo

Why It Is Production-Ready:

- Combines domain-specific security (guardrails) with strict schema enforcement (JSON parsing) to prevent model hallucination and interface breakage.
- Extensible design allows coaches to update the knowledge base or drop in custom PDF training schedules dynamically.

Live Demo Overview:

- Demonstrate custom PDF RAG retrieval, 800m track interval generation in the chat, and JSON-parsed nutrition schedules.


Thank you